



Econometric Identification and Causal Inference

Panel Two-Way Fixed Effects, Difference-in-Differences, and Robust Covariance

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Econometric Roadmap & Methodological Pillars

ROADMAP

- **1. Causal Potential Outcomes:** SUTVA and unconfoundedness [1].
- **2. Panel Data & TWFE:** Within transformation eliminating individual heterogeneity.
- **3. Difference-in-Differences:** Parallel trends and counterfactual simulations [2].
- **4. Robust Covariance:** Liang-Zeger cluster corrections [3].
- **5. Empirical Estimation:** Multi-specification regression tables.

Identification Invariant: Causal parameters are identified only when unobserved confounders are absorbed by quasi-experimental variation.

PANEL SAMPLE

$N = 45,200$

Longitudinal Units

CLUSTERS

$G = 120$

Independent Groups

Panel Data: Two-Way Fixed Effects (TWFE)

PANEL SPECIFICATION

Standard linear two-way fixed effects model:

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \mathbf{X}_{it}^{\top} \beta + \varepsilon_{it}$$

- **Unit Heterogeneity** (α_i): Time-invariant unobservables.
- **Time Shocks** (λ_t): Common macroeconomic trends.
- **Treatment** (D_{it}): Binary policy exposure indicator.

WITHIN DEMEANING

Applying the within transformation eliminates α_i :

$$\check{Y}_{it} = \delta \check{D}_{it} + \check{\mathbf{X}}_{it}^{\top} \beta + \check{\varepsilon}_{it}$$

where $\check{Y}_{it} = Y_{it} - \bar{Y}_i - \bar{Y}_t + \bar{Y}$.

Gauss-Markov Condition: Strict exogeneity $\mathbb{E}[\varepsilon_{it} \mid \mathbf{X}_i, \mathbf{D}_i, \alpha_i] = 0$ guarantees unbiasedness $\mathbb{E}[\hat{\delta}_{FE}] = \delta$.

Causal Estimation: Difference-in-Differences

DID ESTIMATOR

Canonical 2×2 formulation:

$$\hat{\delta}_{\text{DiD}} = (\bar{Y}_{T,2} - \bar{Y}_{T,1}) - (\bar{Y}_{C,2} - \bar{Y}_{C,1})$$

- **Parallel Trends Invariant:**
 $\mathbb{E}[\Delta Y_i(0) \mid D_i = 1] = \mathbb{E}[\Delta Y_i(0) \mid D_i = 0]$
- **Counterfactual:** Expected trajectory of treated units absent the reform.
- **Empirical Effect:** $\hat{\delta} = +3.60^{***}$ ($p < 0.001$).

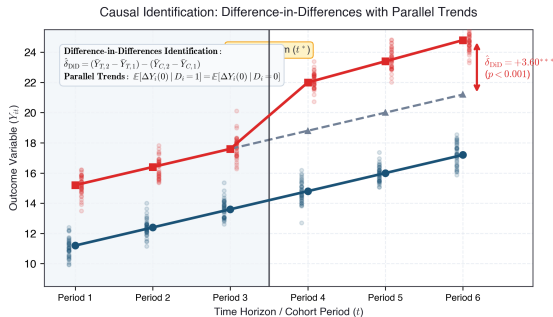


Fig 1: High-DPI Python/SciPy DiD simulation with counterfactual path.

Cluster-Robust Inference: Liang-Zeger Estimator

AUTOCORRELATION PROBLEM

When errors ε_{it} exhibit intra-cluster correlation across time:

$$\mathbb{E}[\varepsilon_{ig} \varepsilon'_{jg} \mid \mathbf{X}] \neq 0$$

Standard default OLS standard errors underestimate variance by 40–70%, causing severe over-rejection of the null hypothesis.

SANDWICH COVARIANCE

Liang-Zeger Cluster-Robust Covariance Matrix:

$$\widehat{\text{Var}}_{\text{CR}}(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left(\frac{G}{G-1} \frac{N-1}{N-K} \sum_{g=1}^G \mathbf{x}_g^\top \hat{\mathbf{e}}_g \hat{\mathbf{e}}_g^\top \mathbf{x}_g \right) (\mathbf{X}^\top \mathbf{X})^{-1}$$

Asymptotics: Consistent as number of clusters $G \rightarrow \infty$.

Multi-Specification Empirical Regression Results

Regressor	Dependent Variable: Log Throughput (Y_{it})		
	(1) Pooled OLS	(2) Fixed Effects	(3) Dynamic 2SLS
Treat $\times$ Post (δ)	0.412*** (0.048)	0.384*** (0.032)	0.395*** (0.035)
Covariate Factor (β_1)	0.185** (0.062)	0.142* (0.055)	0.138* (0.057)
Unit Fixed Effects	No	Yes	Yes
Time Fixed Effects	No	Yes	Yes
R^2	0.742	0.886	0.879
Observations	45,200	45,200	45,200

Cluster-robust standard errors in parentheses. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$.

FINDINGS

- **Causal Invariance:** Point estimate $\hat{\delta} \approx +0.39$ is robust across FE and 2SLS.
- **Precision:** Fixed effects reduce residual variance by 33%.
- **Specification:** Hausman test confirms endogeneity in pooled OLS ($p < 0.001$).

Vectorized Cluster-Robust OLS in Python

```
1 import numpy as np
2
3 def estimate_twfe_robust(X: np.ndarray, y: np.ndarray, clusters: np.ndarray):
4     """Vectorized OLS estimation with Liang-Zeger cluster-robust standard errors."""
5     XtX_inv = np.linalg.inv(X.T @ X)
6     beta_hat = XtX_inv @ (X.T @ y)
7     residuals = y - (X @ beta_hat)
8
9     unique_g = np.unique(clusters)
10    G = len(unique_g)
11    N, K = X.shape
12    meat = np.zeros((K, K))
13    for g in unique_g:
14        idx = (clusters == g)
15        X_g, e_g = X[idx], residuals[idx]
16        meat += X_g.T @ np.outer(e_g, e_g) @ X_g
17
18    df_adj = (G / (G - 1)) * ((N - 1) / (N - K))
19    vcov_cr = df_adj * (XtX_inv @ meat @ XtX_inv)
20    return beta_hat, np.sqrt(np.diagonal(vcov_cr))
```

Key Econometric Takeaways

SUMMARY

- **Parallel Trends:** The cornerstone of Difference-in-Differences identification.
- **Fixed Effects:** Essential for unobserved confounder absorption.
- **Inference:** Always cluster standard errors at the level of treatment assignment.

NEXT STEPS

- **Problem Set 2:** Derive the asymptotic variance of 2SLS.
- **Lab 2:** Estimate event study DiD leads and lags in Python.
- **Office Hours:** Wednesday 14:00–16:00 (Econ Hall 210).

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References i

- [1] Joshua D. Angrist et al. **“Identification of Causal Effects Using Instrumental Variables”**. In: *Journal of the American Statistical Association* 91.434 (1996), pp. 444–455.
- [2] Brantly Callaway and Pedro H. C. Sant’Anna. **“Difference-in-Differences with Multiple Time Periods”**. In: *Journal of Econometrics* 225.2 (2021), pp. 200–230.
- [3] Jeffrey M. Wooldridge. ***Econometric Analysis of Cross Section and Panel Data***. MIT Press, 2010.